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Laboratoire d'Économie d'Orléans
Collegium DEG
Rue de Blois - BP 26739
45067 ORLÉANS Cedex 2

TÉL | (33) (0)2 38 41 70 37
MAIL | leo@univ-orleans.fr
www.leo-univ-orleans.fr

Sovereign credit rating and resource-backed loans in African countries

Yacouba COULIBALY¹ • Jean-Marc Malambwe KILOLO² • Tanouh Fadiga HACIME³

¹*Université d'Orléans, LEO, 45067 Orléans, France*

²*United Nations Economic Commission for Africa, Ethiopia*

³*Université de Grenoble-Alpes, CREG, 38000 Grenoble, France*

Abstract

During the past two decades, the number of African countries receiving a sovereign credit rating has increased exponentially compared to the previous period. In fact, faced with the drying up of conventional financing and the growing need for development finance, many African countries are turning to the eurobond market, which considers the sovereign credit rating as a credible signal of the sovereign's ability to pay their debt. At the same time, resource-rich African countries have resorted to resource-back loans from other developing nations — essentially China, whose principle is to use the natural resource endowments of the countries as collateral or payment in kind. Using the entropy balancing method, the purpose of this paper is to test whether African sovereigns resorting to resource-backed loans have on average a lower credit score compared to others. After performing a set of robustness checks, the study shows that the resource-backed loan has a negative and statistically significant impact on the sovereign credit rating for African countries. This finding suggests that since eurobonds and resource-backed loans are in competition for the same natural resources, countries that resort to the latter receive a lower credit rating compared to their peers.

Keywords: • Resource-backed loans • Sovereign credit rating • Natural resources
• Entropy balancing • Government debt.

JEL Codes: O13, H81, H63.

Corresponding author: Yacouba COULIBALY/ coulyacou9@gmail.com

1 Introduction

“Emerging markets with a history of repeated defaults often face a threshold of debt intolerance, particularly when the debt is denominated in foreign currency, thereby negatively affecting their credit ratings.”

– Reinhart, Rogoff & Savastano (2003)

During the past two decades, the number of African countries receiving a sovereign credit rating (SCR) has increased exponentially compared to the previous period. In 2024, 33 African sovereigns have been rated at least once, compared to 1994 when South Africa was the only African state to be rated (UNDP, 2023). This change stems from the drying up of conventional financing and the growing need for development finance, which has restricted African countries to raise money from international capital markets. Therefore, sovereign credit ratings have become an essential step in evaluating their creditworthiness. In fact, empirical research has shown that SCR is one of the main drivers of bond issuance (Gelos et al., 2011; Thomas, 2009). Given their heavy reliance on natural resource endowments, the assessment of the creditworthiness of African states is significantly dependent on the revenues of these commodity exports.

To finance their infrastructures, resource-rich African countries have often resorted to resource-back loans (RBLs) from other developing nations¹ – essentially China, whose principle is to use the endowments of natural resources of countries as collateral or payment in kind (Mihalyi et al., 2022 Coulibaly; 2024; 2025). In fact, China’s role in global finance has increased dramatically over the years, becoming the world’s largest official creditor since 2017, before the IMF and the World Bank². Since RBLs are based on opaque terms and lack transparency³, in addition to being ‘over-collateralized’ (i.e: using excessive amounts of collateral) on commodities whose price is volatile, they charge higher interest rates⁴. In fact, Chinese loan contracts have unusual confidentiality clauses that prevent borrowing countries from disclosing the terms or even the existence of the

¹These agreements are also known as resource-for-infrastructure deals. Bo et al. (2024) mention the Addis Ababa-Djibouti Railway, Lagos and Abuja Light Rail Projects in Nigeria and the Standard Gauge Railway in Kenya as notable examples.

²Horn et al. (2021) note that 50% of China’s official lending to developing countries is “hidden” since it is not reported in the usual official debt statistics. This has an important impact on the sustainability of the debt.

³Opacity and lack of transparency of RBLs (see Mihalyi et al., 2020; Coulibaly, 2024; 2025).

⁴On higher interest rates on RBLs (see Landry and Tang, 2024).

debt, in addition to influencing the domestic and foreign policies of debtors. In addition, Chinese lenders⁵ seek to position themselves as senior creditors, through collateral agreements such as revenue accounts managed by lenders⁶ and promises to keep the debt out of collective restructuring ('no Paris Club' clauses) (Gelpern et al., 2023).

The increased access of African countries to this alternative external borrowing, especially after benefiting from debt forgiveness initiatives, has raised concerns about the sustainability and risk of debt, given that the continent's revenue-to-GDP ratio is on average one of the lowest⁷. In fact, RBLs represent an important part of the GDP of countries, which amounts to 200% in the case of Guinea (Mihalyi et al., 2020; Coulibaly, 2025). Following IMF, (2018; 2020), international credit rating agencies (CRAs) have also expressed concern about RBL. For (Moody's, 2018), the reliance on non-Paris Club lending increases fiscal and external vulnerabilities, increasing the probability of SCR downgrades. In addition, some studies have found that countries without investment grade are more likely to have RBL (Landry and Tang, 2024).

In this paper, we argue that given the perception of RBLs as a major public finance risk leading to crippling levels of debt (see Horn et al., 2021; Coulibaly, 2024; Mihalyi et al., 2022), they constitute an important determinant of SCR. As a risk factor in assessing sovereigns' probability of default, the accumulation of RBLs potentially leads to downgrades of the SCR. Several channels illustrate this point. Firstly, and as already mentioned, RBLs are often non-transparent and off-budget, which complicates the monitoring of actual public debt (Mihalyi et al., 2022; Coulibaly, 2024; 2025). This can lead to an underestimation of sovereign liabilities by supervisory institutions (parliaments, IMF, rating agencies). When these liabilities emerge or explode in the wake of a shock (e.g., falling oil prices or mining production delays), the agencies abruptly downgrade ratings. Secondly, RBLs tie debt repayment directly to future commodity production or revenues. This further exposes the country to international price cycles (Arezki and Brückner, 2012). If prices fall, the country finds itself in difficulty in repaying, increasing the risk of default, often immediately taken into account in sovereign ratings. This is what happened when commodity prices plummeted in 2014, and many borrowing countries faced a debt crisis (Mihalyi et al., 2020; Coulibaly, 2024). Third, RBLs are often negotiated opaquely, without public tender or parliamentary validation, and sometimes involve collateral clauses on mining or oil assets (Mihalyi et al., 2022; Coulibaly, 2025). This opacity is detrimental to the country's institutional perception. Rating agencies strongly criticize this type of

⁵Note that around 77% of Chinese loans are issued by two state banks, i.e: China Development Bank and Eximbank (Mihalyi et al., 2020).

⁶The confiscation of strategic assets is an option in the event of non-payment of loans, which has been termed "debt trap diplomacy" (Horn et al., 2021; Muruko-Jaezuruka and Birks, 2024).

⁷The continent's average tax revenue to GDP was estimated below 16 per cent (UNECA, 2019).

governance as it increases legal uncertainties and the risk of state capture. Fourth, RBLs require repayment in kind or in future revenues, which reduces budgetary leeway. Even if revenues fall, the debt must be serviced (by oil deliveries or cash transfers). This rigidifies public spending and accentuates adjustments in times of crisis, increasing social and political risk — factors taken into account in ratings ([Bova et al., 2014](#)). Finally, investor perception plays a key role. In fact, the use of RBLs is often interpreted by agencies and investors as a signal of weakness: either an inability to access conventional sovereign debt markets or a desire to avoid the transparency requirements demanded by multilateral lenders. This negative signal alters market confidence and leads to an anticipated deterioration in sovereign ratings.

Based on the transmission channels from RBL to SCR discussed above, we therefore assume that African sovereigns resorting to RBL have on average a lower credit score compared to their peers. To test this hypothesis, we used the entropy balancing method. After performing a set of robustness checks, this study shows that RBL has a negative and statistically significant impact on SCR for African countries. This finding suggests that since eurobonds and RBL are in competition for the same natural resources, countries that resort to the latter receive a lower credit rating compared to their peers.

This paper is related to two strands of literature. First, this research belongs to the economic literature on the drivers of sovereign credit ratings. As such, it is closely related to [Ioannou et al. \(2024\)](#) who analyzed 132 countries during the 2000-2017 period and found that Chinese foreign investment under the Belt and Road Initiative (BRI) has a negative effect on the SCR of recipient countries by S&P and Moody's, when officially participating in the BRI program. More generally, this research belongs to the literature on the drivers of sovereign credit rating, as explained in the literature review below, by integrating the role of resource-backed loans in SCR, it also belongs to the narrow literature that examines the impacts of RBL on the ecological efficiency of human development ([Coulibaly et al., 2022](#)) or deforestation (see [Coulibaly, 2025](#)).

The remainder of the paper is organized as follows. Section 2 presents a review of the literature on SCR and RBL. Section 3 reports some stylized facts. Then, Section 4 presents the data and some descriptive statistics. In Section 5, we present the empirical strategy adopted. The main results are presented in Section 6, while Sections 7 and 8 present the sensitivity of these results and explore heterogeneity tests, respectively. Section 9 concludes with the main policy recommendations.

2 Literature review: drivers of sovereign credit rating

This section examines the existing interactions between natural resources, government debt, and sovereign credit ratings, paying particular attention to the crucial role of transparency and institutional quality in the context of resource-backed loans.

2.1 The determinants of sovereign credit ratings

We conduct an in-depth analysis aimed at providing a nuanced understanding of the challenges and opportunities faced by resource-rich developing countries in their quest to secure finance for development, notably through getting a sound credit rating that signals their capacity to pay back their sovereign debt.

The existing literature on the determinants of sovereign credit ratings is extensive and primarily focuses on factors likely to influence the assessments of rating agencies such as Moody's, Standard & Poor's, and Fitch. These include macroeconomic indicators (GDP, GDP per capita, inflation, global risk, propensity to invest), public finances (debt, fiscal balance, budget deficit), monetary and external factors (real interest, reserves, import and export) and qualitative indicators (level of economic development, history default, crisis indicator) that assess a country's ability and willingness to honor its current and future debt obligations on time ([Cheng et al., 2021](#)). [Stawasz-Grabowska \(2020\)](#) uses an OLS model to study the determinants of SCRs in the context of EU countries during the period 2000 to 2011. The author finds that the economic development indicator, the ratio of export over GDP and money supply over GDP, and the ratio of foreign reserves over GDP positively influence credit ratings, while the default history, growth rate of real exchange rate and GDP deflator and while the ratio of external debt over GDP reduce them. In the same European context, [Oliveira et al. \(2012\)](#) focus on 7 countries over the period 2000-2010. By analyzing the determinants under two periods using a multifactor HJM-Gaussian term structure model, they find that during the 'precrisis period', stock market returns, and interest rate sensitive variables are the most important determinants of credit spreads, while during the 'crisis period', the risk of European volatility, macroeconomic variables linked to credit (level of public debt, current account deficit, public investment, and state of the economic cycle) and the inflation rate appear to play an important role. Another significant study in the euro-zone is that of [Basurto et al. \(2010\)](#), which examines the factors influencing the dynamics of sovereign bond spreads in a sample of 10 countries between 2005 and 2010 using an econometric model based on spread equations. Their results suggest that public debt and deteriorating fiscal balances tend to increase sovereign

spreads, which can influence the perception of a country's creditworthiness and therefore potentially affect its sovereign credit ratings.

[Afonso et al. \(2011\)](#), in a study that covered 130 countries over a 10-year period (1995-2005) and using linear and ordered response models. They find that factors such as GDP per capita, GDP growth, public debt, and external debt have a short-term impact on credit ratings, while external reserves, government efficiency, EU membership, and sovereign default influence sovereign credit ratings in the long term. While [Afonso et al. \(2011\)](#) identify a short-term impact of public debt, the work of [Teixeira et al. \(2018\)](#) suggests a lasting influence of political factors such as corruption. Using a probit ordered model, [Teixeira et al. \(2018\)](#) examined the determinants of sovereign credit ratings in a sample of 86 countries for the period 1993-2013. Their results reveal that GDP per capita, GDP growth and investment improve sovereign credit ratings, while unemployment, inflation, external debt, corruption, political instability, and default history worsen them. They also point out that credit ratings tend to fall in times of crisis. In another study, also based on a large sample of 129 countries during the period 1990-2006, [Hill et al. \(2010\)](#) used an ordered probit model to examine the determinants of sovereign credit ratings from the point of view of the three rating agencies, namely S&P, Fitch and Moody's. Their results show that six variables (GDP per capita, GDP growth and its square, debt history, institutional investor rating, and risk premium) are significant for all three agencies. Their results show that six variables (GDP per capita, GDP growth and its square, debt history, institutional investor rating, and risk premium) are significant for all three agencies, while four others (external balance, external debt, inflation, and budget balance) are significant for only two, suggesting heterogeneity between the agencies. In another study, this time on a restricted sample of 26 countries, [Remolona et al. \(2008\)](#) examine the dynamics of sovereign risk between 1990 and 2005. To do this, the authors used a dynamic econometric model. Their results suggest that nominal GDP, GDP per capita, inflation, the external debt/GDP ratio, the current account/GDP ratio as well as qualitative factors such as default history and political risk. Therefore, it appears that the determinants of sovereign credit ratings are both multiple and interdependent, with weightings varying according to geographical and temporal contexts.

2.2 Theoretical framework: natural resources and sovereign credit ratings

The relationship between natural resource abundance and sovereign credit ratings is theoretically ambivalent and highly dependent on the institutional and macroeconomic context. On the one hand, several authors point out that natural resources can improve a state's creditworthiness by increasing its tax revenues and constituting a strategic asset

that can be valued. For example, [Manzano and Rigobon \(2001\)](#) show that this wealth can provide access to more favorable borrowing terms, while [Wright et al. \(2015\)](#) indicate that resource rents provide a useful financial base, particularly for countries with low fiscal capacity. However, this optimistic outlook is largely nuanced by theories linked to the “resource curse”. Indeed, according to [Brunnschweiler and Bulte \(2008\)](#), it’s not wealth itself that’s the problem, but its combination with failing institutions. Similarly, [Mehlum et al. \(2006\)](#) argue that in countries with weak governance, resource abundance exacerbates corruption and clientelism, undermining fiscal stability and creditor confidence. The high volatility of commodity prices is another factor that weakens credit profiles. [De V. Cavalcanti et al. \(2015\)](#) and [Arezki and Ismail \(2013\)](#) highlight that exposure to price shocks, particularly in the oil sector, increases macroeconomic imbalances and makes public finances more uncertain. However, some works insist on the moderating role of institutional quality. [Ploeg \(2011\)](#) argues that prudent policies — such as the creation of sovereign wealth funds or the adoption of counter-cyclical fiscal rules — can transform natural rent into a lever for stability. Furthermore, [Bhattacharyya and Hodler \(2010\)](#) show that the negative effects of resources manifest themselves only in contexts of poor governance. Consequently, rating agencies include in their assessments not only the economic potential of resources but also the risks associated with their management. [Afonso et al. \(2011\)](#) point out that export concentration, corruption, and fiscal instability are among the main factors in sovereign rating downgrades. So, although natural resources can theoretically strengthen states’ repayment capacity, their actual effect on credit ratings depends closely on governance. Using a sample of 87 emerging and developing countries during the period 1984-2018, [Khan et al. \(2020\)](#) confirm this observation of the ‘curse of natural resources’, while placing particular emphasis on the quality of institutions. Their results, obtained using a probit model, stipulate that good institutional quality and good governance could nonetheless transform the potential curse into a blessing, optimizing countries’ credit rating prospects. Furthermore, [Hill Clarvis et al. \(2014\)](#) suggest that good natural resource management could be a determining factor in the assessment of sovereign credit ratings.

2.3 Public debt and sovereign rating: analytical framework

Public debt is one of the fundamental factors considered by rating agencies when assessing sovereign risk. Indeed, the higher a country’s level of indebtedness relative to the size of its economy, the greater the perceived probability of default, which generally leads to a lower rating. This link is well documented in the academic literature.

Firstly, the debt-to-GDP ratio is a key measure of fiscal sustainability. Reflects a country’s ability to repay its obligations without resorting to extreme fiscal adjustments. Thus, according to [Afonso et al. \(2011\)](#), rating agencies systematically assign lower ratings

to countries with high public debt ratios, as this situation is associated with greater uncertainty over public finances in the medium term. Their empirical study, covering several OECD countries, shows that public debt has a significant negative effect on ratings, even after controlling for institutional and macroeconomic factors. Moreover, the composition of the debt is as important as its overall level. Debt denominated in foreign currencies or contracted with external creditors increases exposure to foreign exchange and refinancing risks. As [Mendoza and Ostry \(2008\)](#) point out, a high proportion of external debt increases a country's vulnerability to exogenous shocks, often resulting in a downgrade of its sovereign rating. In times of financial volatility, these countries are more likely to face a loss of market access or a debt spiral. Furthermore, rating agencies take into account the dynamics of debt rather than its static level. This means that a country can maintain a good rating despite a high debt if its growth prospects are solid, its public finances are under control, and if it implements a credible fiscal policy. In this context, [Taylor and Woodford \(1999\)](#) highlight the importance of the differential between the interest rate (r) and the growth rate (g): if $r > g$, debt becomes more difficult to stabilize, increasing the risk of rating downgrade. Some authors have pointed the finger at the heterogeneity of tolerance between countries. In fact, the most advanced economies are often able to maintain higher debt levels than developing countries without suffering an immediate rating downgrade. This distinction is based on the credibility of institutions, the depth of domestic financial markets, and political stability. [Cantor and Packer \(1996\)](#) confirm that the agencies apply a differentiated assessment according to the level of development of a country, which introduces heterogeneity in the sensitivity of the ratings to debt levels.

Our study comes closest to this literature, which analyzes the link between government debt and sovereign credit ratings. However, we are more specifically interested in resource-backed loans and attempt to link them to sovereign credit ratings, given that resource-backed loans are a particular form of external debt. In practice, the mechanism by which resource-backed loans affect sovereign credit ratings is different from the mechanisms by which other non-resource-linked public debts affect credit ratings. Indeed, resource-backed loans oblige borrowing countries to repay their debt contracted with natural resources or the revenues derived from the sale of these natural resources, amplifying the dependence on natural resources which, by their procyclical nature, affect sovereign credit rating, whereas for other types of loan, there is no constraint or formal obligation for the government to exploit natural resources to repay the loans. Since resource-backed loans are generally taken out in US dollars, the collapse of natural resource prices could increase the nominal sizes of these loans ([Coulibaly, 2025](#); [Mihalyi et al., 2022](#); [Coulibaly, 2024](#); [Mihalyi et al., 2020](#)), exacerbating the country's public debt situation. Consequently, resource-backed loans could affect the country's sovereign credit rating by exacerbating the country's fiscal

budgetary situation. In the same vein, the fact that resource-backed loans are linked to commodity prices can have a strong influence on the sovereign credit rating of borrowing countries, for a number of reasons.

Firstly, as the prices of natural resources (oil, gas, minerals, etc.) are highly volatile, countries whose budgets depend heavily on these revenues will be highly exposed to exogenous shocks. This volatility leads to unstable public revenues and makes medium-term budget planning difficult. According to [De V. Cavalcanti et al. \(2015\)](#), resource-rich countries suffer from more pronounced economic cycles, resulting in a greater perception of risk of default, especially during periods of falling prices. Rating agencies incorporate these elements into their models, lowering credit ratings when they believe that a country lacks the mechanisms to smooth these fluctuations. Secondly, the volatility of commodity prices can lead to pro-cyclical fiscal policy. This procyclicality can reinforce imprudent fiscal behavior: in boom times, some governments increase public spending sharply, but without building up reserves for contractionary phases. This dynamic increases financial vulnerability in the event of a downturn. [Arezki and Ismail \(2013\)](#) highlight this fiscal asymmetry: states react little to price increases in terms of savings but then have to resort to debt when prices fall. The agencies, perceiving this imbalance, adjust their ratings downwards to reflect the increased risk of refinancing. Third, procyclicality is compounded by a lack of economic diversification. In many resource-exporting countries, the economy is heavily dependent on one or two commodities. This reinforces the correlation between international price cycles and domestic macroeconomic performance. [Afonso et al. \(2011\)](#) note that countries with a high concentration of exports in commodities tend to have more volatile, or even lower, sovereign ratings than those with a more diversified economic base. Lastly, rating agencies are often procyclical themselves: they may raise ratings in boom times (thanks to temporary improvements in fiscal indicators), then rapidly lower them in times of crisis, thus helping to amplify the volatility of borrowing costs.

Resource-backed loans can deteriorate the sustainability of the country's debt ([Horn et al., 2021](#)), which is likely to negatively affect economic activity (GDP) ([Reinhart and Rogoff, 2009](#)) and reduce sovereign credit ratings ([Coulibaly, 2024](#)).

3 Stylized facts

This section presents the evolution of sovereign credit ratings, differentiating countries that have signed a RBL from those that have not. As shown in Table A.1 in the Appendix, this unconditional correlation between the two variables is negative and significant at the 5% level (see Appendix). Given this result, we can confidently analyze the disparities between the two groups in two stages.

Firstly, we compare the average in RBL and non-RBL countries, respectively; the left panel of Figure 1 shows that, on average, RBL countries have a lower credit score than non-RBL countries; in other words, they are perceived to be more likely to default on their sovereign debt than their counterparts (an average score 2 for RBL countries versus 4 for non-RBL countries). Besides, the right panel of Figure 1 compares the SCR before and after the adoption of resource-backed loans in signatory countries; it shows a lower average SCR after the adoption of resource-backed loans. This means that the perceived risk of default of African countries increases after contracting in RBLs. It is worth noting that the deterioration of the credit score of signatories is so bad that their average score, which was initially higher than non-RBLs, becomes significantly lower. The fact that the RBL countries initially had a higher credit score than their counterparts probably stems from the fact that they are mostly resource-rich countries with predictable financial streams from their commodity sales. As such, they have a higher capacity to meet their payment obligations that are due and are thus perceived as less risky. By committing their natural resources and/or revenue streams from commodity sales, African sovereigns lose control over them and by doing so reduce their capacity to repay other debts. Therefore, credit rating agencies react to this by revising downward their country risk assessment, which results in a worse SCR. It should be noted that Figure 1 depicts average sovereign credit ratings. Secondly, we therefore perform our analysis with the median, as an analysis based on the average can be misleading. Figure 2 reveals a difference in SCR between RBL and non-RBL countries, as the former exhibit a median (the middle line of the box), 25th (bottom hinge of the box), and 75th (top hinge of the box) percentiles of sovereign credit rating lower than the latter. While these figures exhibit a difference between RBL and non-RBL countries, they cannot assess their magnitude or any significance.

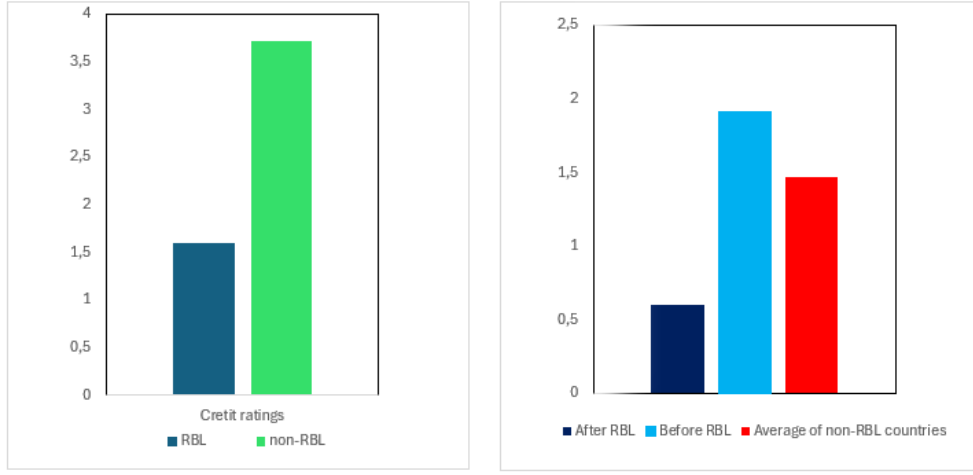


Figure 1: *Sovereign credit rating and resource-backed loans*

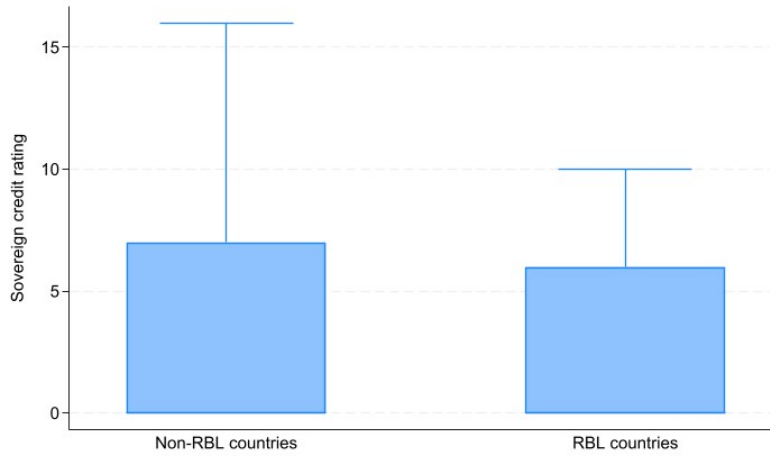


Figure 2: *Sovereign credit rating by resource-backed loans adoption. Note: In box plots, the lower and upper hinges of each box show the 25th and 75th percentiles of the samples, the line in the box indicates the respective medians, and the end-points of whiskers mark next adjacent values.*

Table 1 presents the mean comparison tests between the two groupings of countries. Panel A reveals that the adoption of resource-backed loans appears to reduce SCRs, with a significant difference of 1 percentage point. Panel B shows a greater decline in the average SCR within countries by calculating the average SCR before and after the adoption of resource-backed loans, of 1.89 versus 1 percentage point. Although not causal, these relationships provide an idea of the treatment effect of resource-backed loans and how to identify them. Due to potential secular trends, estimating the effect of resource-backed loans on sovereign credit ratings simply by comparing sovereign credit ratings before and after resource-backed loan adoption is not sufficient. To avoid overestimating the policy effect, we use countries without resource-backed loans as a control group to estimate the counterfactual result.

Table 1: Sovereign credit rating and resource-backed loans: mean-comparison tests

	RBL	Non-RBL	Diff	T-test	P-value
Panel A: Sovereign credit rating	2.32	3.39	1.07	5.3	0.021
Panel B: RBL adoption	Before	After	Diff	T-test	P-value
Sovereign credit rating	2.32	0.43	-1.89	18.4	0.000

4 Data and Descriptive Statistics

4.1 Data description

We examine the effect of resource-backed loans on sovereign credit ratings, using a panel of 45 African countries from 1990-2020. Of this sample, 11 countries signed a resource-backed loan for at least one year between 1990 and 2020. As data are not available for all countries and all years, the number of observations depends on the explanatory variables used in the study.

4.1.1 Treatment variable

The treatment variable used in this study is the adoption of RBL, a binary variable constructed on the basis of information contained in the NRG (Natural Resources Governance Institute) database. To date, the NRG remains the only comprehensive source of resource-backed loans in developing countries (Coulibaly, 2024). A team of researchers led by Mihalyi, Adam, and Hwang compiled two original datasets maintained by research institutions dedicated on China’s overseas activities, namely the China-Africa loan dataset produced by the China-Africa Research Initiative (CARI) at Johns Hopkins SAIS, on the one hand, and the China-Latin America financial database of the Inter American Dialogue and Global Development Policy Center at Boston University, on the other hand (Mihalyi et al., 2020). These sources were supplemented with desk research. It is worth noting that RBL data is not fully comprehensive due to either relative opacity of these transactions (Rivetti, 2021), or limited public data availability (Mihalyi et al., 2020). However, this is the best database on resource-backed loans for African and Latin American economies over a fairly long period. Following the example of Coulibaly (2024); Coulibaly et al. (2022); Mihalyi et al. (2022); Mihalyi et al. (2020), we construct the dummy variable by assigning the value 1 to a country i at a time t if it has taken out a loan guaranteed by natural resources and otherwise.

4.1.2 Outcome variable

Credit rating agencies such as Fitch, Moody's, and assign sovereigns a credit score using an alphabetical rating system from AAA to D (see Table A3 in the Appendix) to evaluate their creditworthiness and the risk level of lending to a given country. This rating plays a crucial role in market access by offering a forward-looking view of the ability of a sovereign debt issuer to fulfill financial obligations. Investors rely on these assessments to assess transaction risks and issuer stability. We use their SCRs and transform them into an ordinal rank from 21 to -1, with the 'not rated' assigned a value of 0 (see Table A3). If a country received multiple ratings from a single agency within one year, we averaged all available ratings. Consequently, countries that defaulted might still have ratings above 0 in that year if they were upgraded after resuming payments. This approach is in line with the literature on the drivers of SCR.

4.1.3 Covariates for matching

We estimated the probability of RBL adoption using a logit model with the binary variable RBL as the dependent variable and performed a correlation matrix between these variables and the treatment variable. The purpose of this study is to measure the correlation between the control variables and the probability of adopting RBL. The control variables included structural and institutional indicators. These factors likely explain a country's decision to adopt a RBL and the evolution of its sovereign credit rating. We therefore monitor the endogeneity of the following factors: GDP per capita, inflation rate, commodity price, political stability, resource depletion, and public debt. It is impossible to control for unobserved factors that can affect the probability of signing an RBL. However, control variables allow us to consider known sources of bias. These data come mainly from the World Development Indicators (WDI), International Monetary Fund (IMF), and World Governance Indicators (WGI) databases. Let us return to explaining the variables.

GDP per capita could have a positive effect on sovereign credit ratings. Indeed, GDP per capita is a crucial indicator of a state's economic development and financial capacity (Afonso et al., 2011; Hadzi-Vaskov and Ricci, 2019). More developed economies generally have more stable institutions, which limits high levels of public borrowing and makes them more resilient to external shocks (Afonso et al., 2011; Butler and Fauver, 2006). According to Cantor and Packer (1996), the level of economic development is one of the most important factors in assessing a country's creditworthiness. Aizenman et al. (2013) point out that this variable reflects economic resilience and the ability to service external debt, particularly in resource-dependent economies. Then, we take into account the macroeconomic situation by controlling the inflation rate (see Sibanda et al., 2018). This variable measures the consumer price index. Its impact on sovereign credit risk is nuanced (Afonso

et al., 2011). A positive effect suggests that inflation can reduce the real stock of public debt in the domestic currency, freeing up resources to cover external obligations (Canuto et al., 2012; Afonso et al., 2011). Conversely, a negative effect suggests that high or volatile rates signal macroeconomic dysfunction (Butler and Fauver, 2006; Sibanda et al., 2018; Hadzi-Vaskov and Ricci, 2019; Erdem and Varli, 2014). Indeed, Dell’Erba et al. (2013) show that inflation exacerbates the procyclicality of credit ratings, particularly in times of crisis. These rates are correlated with accumulated credit risk and lower sovereign ratings (Afonso et al., 2011). In the African context, fluctuating inflation can reduce a country’s repayment capacity and jeopardize access to advantageous lending terms, which could encourage the use of resource-backed loans. Moreover, the literature indicates that African countries, particularly those rich in resources, are vulnerable to fluctuations in commodity prices (Mpapalika and Malikane, 2019). These fluctuations create large-scale economic uncertainties (Gelos et al., 2011), strongly impacting their public finances and sovereign credit ratings. We take into account that observation by controlling the commodity price. According to Mpapalika and Malikane (2019), the influence of commodity prices on credit ratings is ambivalent. An increase in commodity prices can increase export revenues and improve perception of sovereign risk, while a decrease may reduce revenues and increase default risk (Cheuathonghua et al., 2022). Furthermore, they note that the more volatile commodity prices are, the wider sovereign credit default spreads (CDS) increase, reflecting an increase in the perception of sovereign default risk. This indicates that investors believe that countries whose economies rely heavily on commodities are more likely to face financial problems in the event of price fluctuations, which is reflected in government bond yield spreads.

From a theoretical point of view, and especially when analyzing sovereign credit ratings, it makes sense to take into account not only macroeconomic fundamentals, but also institutional fundamentals and uncertainty. In fact, the greater the political uncertainty, the lower the credit rating. Conversely, strong political institutions oblige the government to make a credible commitment not to engage in behavior that harms investors’ interests, and to implement policies and regulations conducive to the country’s growth and stability. For instance, Carmignani (2003) shows that political instability leads not only to political and policy uncertainty, but also to the stability of economic institutions, which all affect agent investment decisions. Consequently, we control the political stability variable in our baseline model. In the spirit of Hadzi-Vaskov and Ricci (2019), high levels of public debt reduce the likelihood of a high sovereign credit rating. At the same time, Coulibaly (2025) argued that high public debt is associated with high likelihood signing of resource-backed loans. Moreover, Reusens and Croux (2017) show that countries with high public debt relative to GDP often have lower credit ratings. Mellios and Paget-Blanc (2006) also

confirm the crucial importance of public debt in the evaluation of financial risk, showing its systematic negative impact on sovereign credit ratings. As a result, we control the effect of public debt in our model.

Finally, we control resource depletion, which refers to a situation in which natural resources are being consumed faster than they can be renewed ([World Bank, 2024](#)). This variable measures the accelerated depletion of non-renewable natural resources such as forests, mineral resources (tin, gold, lead, zinc, iron, copper, nickel, silver, bauxite and phosphate), and energy resources (oil, natural gas and coal). [Nawaz et al. \(2019\)](#) argue that the depletion of natural resources is contributing to reduced economic growth in many countries, which could negatively impact their sovereign credit ratings.

4.2 Descriptive statistics

During the 1990-2019 period, sub-Saharan African countries gradually gained access to international capital markets, notably through the issuance of sovereign bonds in foreign currencies. This movement has been accompanied by a rise in sovereign credit ratings, provided by the main agencies (S&P, Moody's, Fitch) and, in some cases, relayed by institutions such as the World Bank. However, these dynamic changes remain marked by structural asymmetry: in 2019, more 80% of the countries rated in the region were classified in the speculative category (non investment grade), with a concentration of ratings at levels B and B-. Access to ratings accelerated from the 2000s onward; this timing corresponds to the program under the HIPC Initiative and the growing number of African Eurobond issues. Nevertheless, this dynamic has proven fragile: several countries, such as Ghana, Mozambique, Zambia, and Congo-Brazzaville, have experienced episodes of default or restructuring, resulting in successive rating downgrades. Criteria explaining this vulnerability include excessive dependence on raw materials, rising debt levels, persistent budget deficits, and weak governance institutions.

A comparative graphical analysis (see [Figure 3](#) above) reinforces this reading. There is a clear and growing divergence between countries with and without RBLs: Although the ratings of non-RBL countries follow a relatively stable upward trajectory between 2000 and 2019, those of RBL countries stagnate at a lower level, even with a temporary decline between 2015 and 2017. A noticeable drop can be seen around 2014-2015, a period marked by a brutal exogenous shock to commodity prices, particularly oil, which highlighted the vulnerability of states dependent on RBLs. These instruments, based on assumptions of stable future export revenues, proved to be rigid and exposed to global volatility, exacerbating fiscal imbalances and leading to downgrades by rating agencies. This visual trend suggests that markets and rating agencies perceive the use of RBLs as an aggravating factor in sovereign risk, possibly due to the uncertainties they introduce

about transparency, debt sustainability, and commodity dependence.

This figure shows that, although sovereign ratings became progressively more widespread in sub-Saharan Africa between 1990 and 2019, the region remains globally perceived as high-risk by markets. However, fiscal credibility varies from one country to another. This observation motivates the analysis of innovative financing instruments, such as resource-backed loans, and their dynamic influence on sovereign risk perception in the region.

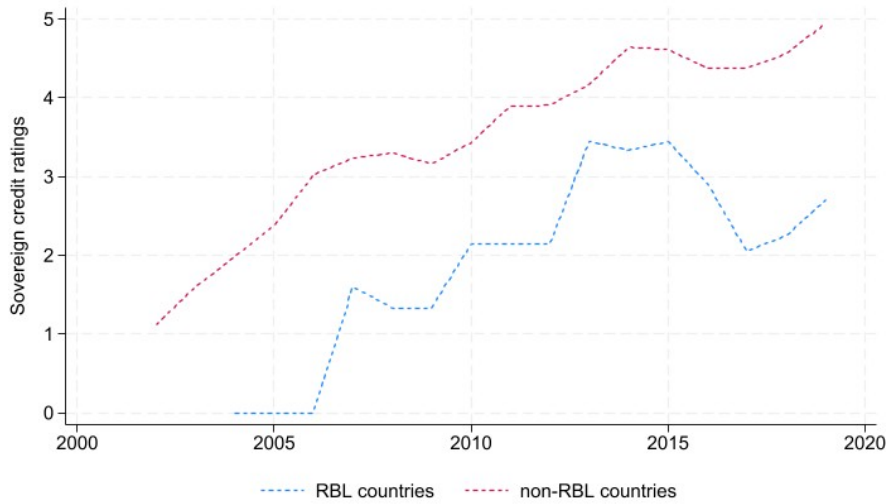


Figure 3: *Average of sovereign credit rating by resource-backed loan adoption over period 2000-2020*

We use the method of [De Chaisemartin and d'Haultfoeuille \(2024\)](#) to capture the heterogeneous effects of resource-backed loans over time. We capture trends in credit ratings a few years before the adoption of resource-backed loans and years after adoption. We choose a range of two years before and five years after the introduction of resource-backed loans. Figure 4 shows the results. First, we observe no pre-trends before adopting resource-backed loans. This evidence clearly indicates a parallel trend between the two groups (RBL and non-RBL countries) because the effect is statistically insignificant in the episodes preceding adoption, even though the time dimension seems quite small. The estimated credit rating coefficients are all positive and insignificant during these periods. After adopting resource-backed loans, we observe that the coefficient of the credit rating share becomes negative from year 1 to year 5. The negative effect is significant from the first year after adoption and extends to the fifth year, after which the effect becomes positive but insignificant. The significant effect of resource-backed loans on countries' public finances in terms of worsening budget balances and public debt sustainability can be explained by the speed with which they are achieved. Another explanation could be the lack of transparency regarding resource-backed loans, especially information about such loans. In fact, countries that subscribe to RBLs often conceal the fact that they

have signed an RBL with China or elsewhere because they do not want to send out the wrong signal on the financial market. This could prevent creditor countries from assessing the creditworthiness of the debtor country before granting it a loan that is not a resource-backed loan (see [Coulibaly, 2024](#)). Finally, it should be noted that the average of these coefficients over the 6 years following the adoption of the resource-backed loan is -0.55 percentage points.

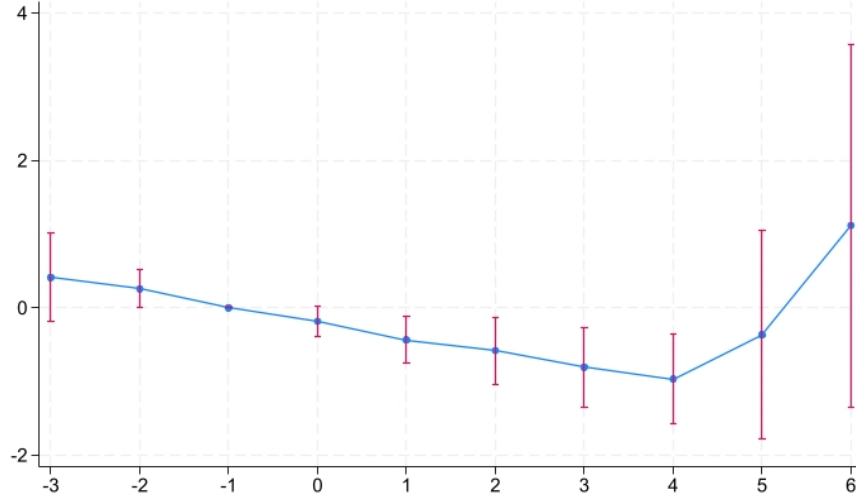


Figure 4: *Dynamic effects of RBL on credit ratings applying the dynamic heterogeneity robust difference-in-difference estimator of [De Chaisemartin and d’Haultfoeuille \(2024\)](#). Bars represent 95% confidence intervals around the effects of long-difference tests, the number of placebo (2) and 50 bootstrap replications were applied.*

5 Methodology

The main objective of this study is to analyze the influence of resource-backed loans (RBL) on African sovereign credit ratings ([Coulibaly, 2024](#); [Mihalyi et al., 2020](#); [Mihalyi et al., 2022](#)). As an alternative form of financing subject to macroeconomic and institutional risks, such as the risk of sovereign default or debt restructuring, limited access to financial markets, and increased corruption, RBLs raise concerns about potential selection bias that needs to be addressed. Indeed, the RBL variable may be correlated with unobservable variables affecting the overall performance of the economy, but also the sovereign risk measured by rating agencies. Analyzing the impact of resource-backed loans on sovereign credit rating is particularly important in the current African context, due to their limited access to financial markets.

To perform this analysis, we chose to apply an entropy balancing method developed by [Hainmueller \(2012\)](#) which has been widely used in the literature ([Balima et al., 2017](#);

Kinda and Thiombiano, 2024; Coulibaly, 2024), is based on a binary treatment where observations require covariates adjustments to capture causal effects. This approach. In addition, it is recognized for its ability to minimize selection bias and provide robust estimates. In this study, we treat our approach as based on the principle that the adoption of resource-backed loans is the treatment, and the sovereign credit rating is the outcome variable. The units of observation are countries in year t . Observations that have adopted RBL constitute the treatment group, while those without RBL constitute the control group as described above. The effect of treatment on treated individuals (ATT) is defined as follows:

$$ATT = E[(Y_{(1)}|T=1, X = 1)] - E[Y_0|T=1, X = 1] \quad (1)$$

where $Y_{i(.)}$ denotes the sovereign credit rating. T_i indicates whether the observation unit is subject to the adoption of resource-backed loans ($T_i = 1$) or not ($T_i = 0$). $E[(Y_{i(1)}|T_i=1)]$ is the level of sovereign credit rating during the resource-backed loan adoption period, $E[(Y_{i0}|T_i=1)]$ is the counterfactual result for countries that had adopted resource-backed loans, i.e., the sovereign credit rating in countries that had adopted resource-backed loans if they had not. The issue is that $E[Y(0)|T = 1]$ is not observable due to the non-random nature of the adoption of resource-backed loans. If this were the case, the ATT could easily be identified by comparing the sovereign credit rating in resource-backed loans countries with non-resource-backed loans countries. Identifying ATT then requires a good proxy for $E[Y(0)|T = 1]$. To do so, we match resource-backed loans units with non resource-backed loans units (after purging for some specific factors) that are as close as possible on observable characteristics that meet two criteria: correlated with RBL adoption and sovereign credit rating. Under the condition that the non resource-backed loans units are fairly close to the resource-backed loans units, any difference in sovereign credit rating is attributable to resource-backed loans adoption. Based on these details, we can rewrite Equation 1 to get Equation 2 as follows:

$$ATT = E[(Y_{(1)}|T=1, X = x)] - E[Y_0|T=1, X = x] \quad (2)$$

where $X = x$ is a vector of observable covariates that can affect both the decision to adopt resource-backed loans and the sovereign credit rating, $E[Y(1)|T = 1, X = x]$ is the sovereign credit rating of resource-backed loan units and $E[Y(0)|T = 0, X = x]$ is the expected sovereign credit rating for synthetic control units. The entropy-balancing approach is a two-stage estimation method. First, the weights assigned to the control units (in this case, the non-RBL countries) are calculated. The weights obtained in the first step are then used in a regression analysis with the treatment variable in this case ‘countries with RBL’ as the explanatory variable. Then, on the basis of observable characteristics, the

treated (RBL countries) and controls (non-RBL countries) are weighted. Consequently, the average difference in sovereign credit ratings between countries with RBL and the ‘closest’ countries without RBL should be explained by the adoption of RBL.

Unlike other methods of estimating treatment effects, such as propensity score matching or difference-in-difference, entropy balancing offers several advantages due to its ability to combine matching and regression analysis (Hainmueller, 2012; Neuenkirch and Neumeier, 2016; Sawadogo, 2020; Balima et al., 2017; Coulibaly, 2024). Indeed, this approach proves preferable to other classical matching methods based on regression and propensity score methods because of its non-parametric nature, which circumvents the problem that a poor specification of the functional form of the model can lead to biased results. Entropy balancing has the advantage of using more flexible reweighting schemes compared to conventional balancing, where control units can be rejected. Since the objective is to balance treated and untreated individuals, keeping the equilibrium weights as close as possible to the initial weights avoids any loss of information. Indeed, its main feature lies in its ability to balance covariates to a high degree between RBL and non-RBL countries, even in a small sample, by creating a synthetic control group as close as possible to the program group. In addition, this method eliminates multicollinearity problems, since the reweighting mechanism makes the treatment variable orthogonal to the covariates (Hainmueller, 2012).

The use of entropy balancing also makes it possible to take account of the dimension of the panel data by controlling for country- and period-specific factors in the regression analysis (second stage). In order to account for potential unobserved heterogeneity between RBL and non-RBL countries, it is therefore necessary to include country-specific effects. This is done under the pretext that the macroeconomic environment between these two groups of countries may differ according to the covariates used in the entropy balancing. However, this approach cannot control for some potential biases insofar as unobserved endogeneity related to temporal factors may affect country ratings, as well as possible reverse causality problems between treatment variables and outcomes: this could lead to potentially biased results. For this reason, we complement our entropy balancing approach with alternative estimation methods.

6 Results

6.1 Results from the first step (covariates balance)

Table 2 presents the descriptive statistics related to the first-step equation. Panel A compares the sample means before weighting the corresponding covariates described in Section 4.1, between African sovereigns holding those without RBL serving as “control

units” or “potential synthetic group” (column [2]). Column [3] indicates statistically significant differences between the two groups, as some p-values are below the 10% threshold. Specifically, countries with RBL are more likely to have higher inflation and less political stability than their counterparts. Such differences could lead to selection bias in policy adoption and erroneous estimates if endogeneity is not properly addressed. Against this background, we re-weighted the control units to make the pretreatment covariates of the control group, on average, as comparable as possible to those of the treated units. The means of the synthetic group covariates are reported in column [1] of panel B. Column [3] indicates that the weighting has eliminated any statistically significant pre-treatment differences between the means of the treated and synthetic covariates. The synthetic group can therefore be considered an “almost perfect” counterfactual to the treated group, thus resolving potential selection problems due to policy adoption.

Table 2: Covariates balance

	[1]	[2]	[3]=[2]-[1]	[4]	[5]
Panel A : Descriptive statistics before weighting	RBL	Non-RBL	Diff	t-Test	P-value
Lag GDP per capita	2.41	1.75	-0.66	1.5	0.222
Lag inflation rate	13.06	7.96	-5.1	6.5	0.011
Lag public debt	55.36	59.46	4.1	0.0	0.932
Resource depletion	10.50	7.30	-3.2	4.0	0.046
Commodity price	96.83	96.90	0.07	9.5	0.002
Political stability	0.45	0.49	0.04	0.3	0.570
Panel B : Descriptive statistics after weighting	RBL	Non-RBL	Diff	t-Test	P-value
	[1]	[2]	[3]=[2]-[1]	[4]	[5]
Lag GDP per capita	2.408	2.408	0.00	0.2223	0.8241
Lag inflation rate	13.06	13.06	0.00	0.4935	0.6217
Lag public debt	55.36	55.37	0.10	-1.6702	0.9524
Resource depletion	10.5	10.5	0.00	2.4087	0.9919
Commodity price	96.83	96.83	0.00	-4.6413	1.0000
Political stability	0.4505	0.4505	0.00	1.8725	0.9693

Note: Panel A presents the pre-weighting sample means of the matching covariates for country-year observations where RBL were signed (the treatment group) in column [1] and country-year observations where no RBL were signed (the potential control group) in column [2] while panel B presents the sample means matching covariates after weighting across the treated group and the synthetic control group obtained from entropy balancing.

6.2 Main results

The second step in entropy balancing is to estimate the treatment effect (resource-backed loans) based on the weights calculated in panel B of Table 2. We estimate the following model:

$$SCR_{it} = \alpha + \beta RBL_{it} + \gamma X_{it} + \nu_i + \theta_t + \epsilon_{it} \quad (3)$$

where SCR_{it} is the sovereign credit rating of country i in year t . RBL_{it} is a dummy variable equal to 1 for a country i that has adopted a resource-backed loan in year t , and 0 otherwise. X_{it} is the set of covariates described in Section 4.1. The ν_i and θ_t represent country- and time-fixed effects, respectively, capturing unobserved heterogeneity. Finally, ϵ_{it} is the term for idiosyncratic error. The main results are presented in Table 3. In column [1], we run a simple uni-variate regression called “naive regression” from entropy balancing to capture the only sovereign credit rating reactivity following the adoption of resource-backed loans. Column [2] includes all the controls of the baseline model. In columns [3]-[4], we include country and year fixed effects, respectively. Finally, column [5] presents the main estimation considering covariates and both country and year-fixed effects. Estimates suggest that contracting resource-backed loans reduces African sovereign credit rating on average by 2.09 percentage points compared to African countries that don’t hold such loans. This result is statistically significant at the 1% level (Table 3, column [5]). His effect represents approximately 46% of the standard deviation of the outcome variable, suggesting that resource-backed loans have an economically significant impact. Finally, regarding the control variables of the baseline model, we find that resource depletion and inflation rate significantly reduce the sovereign credit rating, whereas public debt, commodity price, and political stability affect it positively⁸. In column [6], we perform a kind of robustness check by analyzing the influence of particular confounding factors that may pollute our effects by including a time trend in the previous model. We refer in particular to [Coulibaly \(2024\)](#), who argued that controlling for time trend eliminates distinctive trends in our outcome variable in individual countries that might otherwise bias our estimates if they accidentally coincided with other changes in resource-backed loans. The results are consistent with those of the basic model.

The negative impact of resource-backed loans on African sovereign credit ratings is intuitively explained by several factors. First, the negative effect could be due to fiscal indiscipline (debt pro-cyclicality). Given that resource-backed loans offer the opportunity to renegotiate their debts and take out another RBL, debtor countries can end up being over-indebted. In the same vein, the grace period that they often enjoy could increase their

⁸A full table with covariates results could be provided to readers on request. We have decided not to report these results in Table 3 due to space constraints.

moral hazard and debt burdens. Second, resource-backed loans suffer from lack of greater transparency (Mihalyi et al., 2020), resulting into over-indebtedness as the debt amounts are not recorded in public finance statistics. Third, given that resource-backed loans are generally denominated in US dollars, a collapse in natural resource prices increases the debt burden, eventually leading to sovereign debt. This is exactly what happened to 10 countries that signed up to resource-backed loans and found themselves in over-indebtedness after the collapse in commodity prices in 2014 (Mihalyi et al., 2022).

In summary, credit rating agencies African countries using RBLs as riskier borrowers, with a tight budget and limited financial resources. This behavior is considered as a negative signal from the debtor country regarding its financial conditions and its ability to repay the debt given the high volatility of natural resource prices.

Table 3: Resource-backed loans and sovereign credit rating—Baseline results

	[1]	[2]	[3]	[4]	[5]	[6]
Variables						
	<i>Naive regression</i>	Adding Controls	Controls & Country/FE	Controls & Year/FE	Controls & Country/Time FE	Controls & Time & Country FE/Trend
RBL adoption	-1.1895*** (0.4500)	-1.1883*** (0.3852)	-1.2801*** (0.4556)	-1.6705*** (0.4484)	-2.0893*** (0.4713)	-2.0893*** (0.4713)
Constant	3.5476*** (0.2011)	3.0595 (5.8689)	-0.8563 (4.2661)	4.8866 (5.5184)	0.0537 (3.8352)	-3.0081 (3.8326)
Observations	639	639	639	639	639	639
R-squared	0.0212	0.2648	0.8126	0.2874	0.8375	0.8375

Note: This table presents the effect of resource-backed loans on sovereign credit rating obtained by weighted least squares regressions. The treatment variable is the presence of a resource-backed loan. The outcome variable is the sovereign credit rating. The control variables include GDP per capita, inflation rate, commodity price, resource depletion, and political stability. Robust standard errors are in parentheses *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7 Robustness

7.1 Alternative estimation methods

In this section, we check whether our baseline results are robust alternative estimation methods, such as OLS (ordinary least squares), PSM (propensity score matching), system GMM and IV (instrumental variables) estimators.

7.1.1 Propensity score matching

We check by applying another impact estimation technique, propensity score matching (PSM). Indeed, the principle of the PSM technique is to compare countries that have signed a resource-backed loan with countries that have not, on the basis of certain observable characteristics linked to both the adoption of RBLs and sovereign credit ratings. These observable characteristics are summarized in a propensity score (PS), which indicates the probability that a country adopts an RBL (treatment group) as a function of the observable characteristics. The PS is then used to identify countries that have not adopted RBL (control group). This control group serves as a counterfactual for the treatment group. Assuming that the determinants of sovereign credit ratings are statistically independent of RBL adoption, given the common characteristics between the treatment and control groups, the difference in outcome between the two groups, the average treatment effect on treated (ATT) is attributable to RBL adoption. PS is estimated using a probit model with a dummy variable for a given RBL as a dependent variable. We use three matching algorithms for country matching to test the robustness of our results. The matching techniques used include radius matching to match an RBL country to non-RBL countries with PS falling within a radius of length r (we consider a wide radius $r = 0.09$ and a narrow radius $r = 0.18$). Then, regression-adjusted local linear matching is used to associate the covariate-adjusted results for the treatment group with the corresponding covariate-adjusted results for the control group using local linear regression weights (Fan, 1993). Finally, kernel matching is used as a matching algorithm to associate a treated country with all control countries weighted proportionally by their proximity in terms of PS to the treated country. The new ATTs in Table 4, confirm the negative effect of RBL on SCR.

Table 4: Resource-backed loans and sovereign credit rating: PSM results

	Radius Matching	Radius Matching	Local Linear	Kernel
VARIABLES	<i>N=0.09</i>	<i>N=0.18</i>	Matching	Matching
RBL adoption	-1.2239** (0.5288)	-1.4615*** (0.4111)	-0.8663** (0.4284)	-0.8728* (0.4985)
Observations	639	639	639	639

Bootstrapped standard errors based on 50 replications are in parentheses. *** $p < 0.01$,

** $p < 0.05$, * $p < 0.1$.

7.1.2 OLS estimates

We re-estimate our main model using a simple fixed-effects panel regression, based on the OLS estimator. The results presented in column [1] of Table 5 suggest a negative and statistically significant effect of resource-backed loans on sovereign credit ratings. Moreover, the effect obtained by OLS (1.7 percentage points) remains qualitatively comparable to that of the basic model obtained by entropy balancing (1.48 percentage points on average).

7.1.3 GMM system and instrumental variable

To further assert, two complementary econometric methods have been used: the system generalized method of moments (system GMM) and the instrumental variable approach (IV). The use of the GMM system, proposed by [Arellano and Bover \(1995\)](#) and [Blundell and Bond \(1998\)](#), allows us to control for biases linked to dynamic endogeneity and unobserved fixed effects by exploiting internal instruments derived from the lags of the explanatory variables. This method is particularly well suited to non-stationary panels with a large number of countries (N) and a limited time horizon (T), as in our sample. The results are presented in column [2] of Table 5. These results lead to qualitatively similar conclusions to the baseline results. In addition, if one looks at the instrument selection criteria, the Hansen test does not reject the hypothesis of instrument validity. Finally, the AR (1) test for the absence of autocorrelation of the first-order error term and the AR (2) test for the absence of autocorrelation of the second-order error term do not raise concerns about the validity of our estimates.

Finally, we adopt a two-stage least squares (2SLS) estimation strategy with instrumental variables. In this framework, we use as instruments the lagged values (lags) of potentially endogenous explanatory variables, including the RBLs themselves. This method is based on the assumption that past values of RBLs are correlated with their current level, but are not directly correlated with the error of the structural regression explaining the current rating, once fixed effects and macroeconomic controls are included. The use of lags as instruments is particularly suitable in the context of time-structured panel data, provided that the errors are not persistently autocorrelated. In addition, instrument validity tests, such as Hansen’s over-identification test and the first-stage F statistic, are carried out to ensure that the lags used are sufficiently correlated with the instrumented variable (relevance) and that they respect the exogeneity assumption (exclusion restriction). The results reported in Table 5, column [3] reveal that the adoption of resource-backed loans significantly reduces the SCR when we take into account the endogeneity problem.

Table 5: Resource-backed loans and SCR: OLS, system GMM and IV-2SLS

	[1]	[2]	[3]
VARIABLES	<i>OLS</i>	<i>GMM</i>	<i>IV-2SLS</i>
Lag.ratings		0.226*** (0.057)	
RBL adoption	-1.710*** (0.411)	-2.026** (0.779)	-1.150*** (0.418)
Lag GDP per capita	0.007 (0.017)	0.002 (0.022)	0.004 (0.038)
Lag Inflation rate	-0.024** (0.011)	-0.008 (0.016)	-0.009 (0.011)
Commodity price	0.097*** (0.029)	-0.018 (0.036)	-0.021 (0.034)
Political stability	7.571*** (1.466)	9.300*** (3.237)	12.635*** (1.251)
Resource depletion	-0.100*** (0.034)	-0.037 (0.035)	-0.058*** (0.019)
Lag Public debt	0.007*** (0.002)	-0.013* (0.007)	-0.017*** (0.004)
Constant	-5.266* (2.819)	1.698 (4.046)	1.474 (3.540)
Number of instruments		27	7
AR(1) test, p-value		0.001	
AR(2) test, p-value		0.387	
Sargan, p-value		0.000	
Hansen, p-value		0.176	0.5329
Observations	639	611	638
R-squared	0.848		0.254
Number of country		38	

Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

7.2 Placebo test

The results of this placebo test reported in Table 6 of RBL on sovereign credit ratings, reinforcing the idea that the correlation between both variables does not stem from anticipation mechanisms or pre-existing structural characteristics. This therefore supports the hypothesis of a delayed causal effect between RBL commitments and the deterioration in sovereign risk profile, by ruling out the possibility of reverse causality bias or omission of temporal variables.

Table 6: Resource-backed loans and SCR: placebo test

	[1]	[2]	[3]	[4]	[5]
VARIABLES					
RBL random	-0.358 (0.389)	-0.340 (0.334)	-0.030 (0.175)	-0.237 (0.340)	0.119 (0.169)
Constant	4.087*** (0.290)	3.058 (3.512)	-6.230** (2.820)	4.165 (3.511)	-5.821** (2.772)
Covariates	No	Yes	Yes	Yes	Yes
Country/FE	No	No	Yes	No	Yes
Year/FE	No	No	No	Yes	Yes
Observations	639	639	639	639	639
R-squared	0.001	0.259	0.822	0.281	0.848

Robust standard errors in parentheses.*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

7.3 Alternative samples

In this section, we carry out a series of robustness tests by modifying the overall sample to ensure the robustness of the results that estimate the impact of resource-backed loans on sovereign credit ratings in developing countries. These re-estimates allow us to identify whether the results are sensitive to certain subpopulations or specific time periods. Firstly, we exclude periods of major macroeconomic shocks, which are likely to distort rating agencies' assessment of sovereign risk, irrespective of debt structure. The following periods have been excluded: the global financial crisis (2008-2009), the oil crisis (2014-2015) and the early post-Cold War years (1990-1995). These periods are characterized by extreme disruptions in international financial flows and atypical rating policies. Secondly, we exclude countries in prolonged armed conflict, where sovereign ratings incorporate extreme geopolitical risks that can overshadow the specific effect of RBLs. Similarly, net oil exporting countries are excluded after assessing the importance of RBLs in the absence of oil dependence. Third, one test consists of restricting the sample to countries that contracted RBLs at least five years before the end of the observation period, to take

account of the time lag in the rating agencies' reactions. This makes it possible to assess the medium-term effects of RBLs on sovereign credibility beyond the immediate adjustments. Fourthly, we exclude the most macroeconomically unstable countries (i.e. those that have experienced hyperinflation), in order to check whether the estimated effect of RBLs persists in contexts of relative stability. Finally, the sample is also re-evaluated after excluding the 5% of extreme observations on debt or rating levels (winsorization), to ensure that the results are not pulled by specific outliers. In general, these sample modification tests establish the empirical robustness of the link between recourse to RBLs and sovereign rating). The results reported in Table 7, column [1]-[8] reveal that the resource-backed loan adoption significantly reduces the SCR when we take into account these modifications.

7.4 Additional controls

To rigorously analyze the impact of resource-backed loans on sovereign credit ratings, it is essential to control for several macroeconomic, institutional, and structural factors likely to affect both the likelihood of resorting to RBLs and the perception of sovereign risk by rating agencies. Capital openness plays a fundamental role in market discipline: countries with greater exposure to international financial flows may come under greater pressure to maintain their fiscal credibility, which has a direct impact on their ratings. International reserves, measured in months of imports (reserves/months), indicate a country's ability to meet external obligations: high reserves are generally viewed positively by rating agencies, as they reflect greater resilience to liquidity shocks. The status of resource-rich countries is a key determinant of RBLs, but also a source of vulnerability if future revenues are anticipated too optimistically, a factor taken into account in sovereign ratings. The exchange rate regime (fixed exchange) determines macroeconomic flexibility: a rigid regime can make adjustment more costly in the event of a crisis, which can amplify the risks associated with RBLs and affect ratings. The type of government, including its political stability and orientation, influences the ability to manage complex loans such as RBLs, particularly in terms of transparency and accountability. The level of governance, measured by corruption control, directly affects institutional credibility and public debt management. In a context of high corruption, rating agencies may consider resource-backed commitments to be particularly risky. Furthermore, a history of external default or financial crises weakens market confidence and increases rating sensitivity to any new form of unconventional debt. Finally, the past occurrence of inflationary crises may signal chronic macroeconomic instability, reinforcing the importance of controlling for these variables in any estimate of the effect of RBLs on sovereign ratings. The results reported in Table 7 panel B, columns [1]-[8] show that resource-backed loans significantly reduce

sovereign credit ratings.

Table 7: Resource-backed loans and sovereign credit rating: sample modification

	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Panel A: Excluding observations	Financial crisis	Oil crisis	Post-cold War	Armed conflict	Net oil-exporting	RBL at least 5-years	Hyperinflation	Outliers
RBL	-2.089*** (0.471)	-2.183*** (0.498)	-1.853*** (0.499)	-0.816** (0.408)	-0.941** (0.407)	-1.734*** (0.563)	-1.709*** (0.588)	-2.089*** (0.471)
Constant	0.054 (3.835)	0.965 (3.037)	-2.665 (4.705)	-9.802*** (3.478)	-1.477 (3.390)	-12.570*** (4.774)	-16.069*** (5.011)	0.054 (3.835)
Observations	639	567	564	455	558	601	593	639
R-squared	0.837	0.850	0.825	0.880	0.863	0.844	0.848	0.837
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE & Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
Panel B : Adding variables	Capital openness	Total reserve	Resource-rich	Exchange regime	Governance stability	Control of corruption	Payment defaults	Inflation crisis
RBL adoption	-2.0939*** (0.4736)	-0.9284* (0.5027)	-1.4829*** (0.4848)	-2.2946*** (0.5143)	-0.8791* (0.4864)	-2.2806*** (0.4836)	-3.0761*** (1.1490)	-2.4172** (1.0204)
Constant	1.3239 (3.9449)	-2.0057 (3.3268)	-13.7930*** (3.1888)	-2.4819 (4.5133)	-3.0723 (3.3472)	-0.8315 (3.9784)	0.7090 (5.6971)	-3.4204 (6.4455)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE & Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	639	432	372	528	420	639	141	141
R-squared	0.8387	0.8967	0.8491	0.8557	0.8611	0.8489	0.8923	0.8824

Robust standard errors in parentheses.*** p<0.01, ** p<0.05, * p<0.1

7.5 Alternative definition of credit rating

Finally, in this section, we analyze the sensitivity of our results to a change in the dependent variable, by replacing sovereign ratings from private agencies (used in the baseline estimation and described in Subsection 4.1.2) with the sovereign debt ratings indicator developed by the World Bank (see Kose et al., 2017). This institutional database offers an alternative, synthetic and harmonized measure of sovereign risk, by aggregating the assessments of the three main rating agencies (Moody's, S&P and Fitch) and ranking them on a unified scale. This change is designed to ensure that the results do not depend on the specific methodology of a particular agency nor on the specifics of their rating cycle.

The results reported in the Table 8 columns [1]-[6] of this alternative specification fully confirm the relationship highlighted in the basic model. The estimated effect of resource-backed loans on the sovereign credit rating remains negative and statistically significant, indicating that these financing instruments - due to their lack of transparency, their dependence on volatile commodity prices, and their contractual rigidity — continue to be perceived as a signal of weakening fiscal position and debt sustainability. This consistency in the sign and significance of the coefficient, despite the change in measurement of the dependent variable, considerably strengthens the empirical credibility of the results. Moreover, the use of the World Bank indicator offers an additional advantage in terms of temporal coverage and inter- country comparability, making it possible to slightly broaden the analysis sample. This stability of results suggests that the observed link between the

use of RBLs and the downgrading of sovereign ratings is not a measurement artifact, but rather a robust structural effect, detectable regardless of the methodological angle used to assess the financial credibility of states.

Table 8: Resource-backed loans and SCR: alternative definition of SCR

	[1]	[2]	[3]	[4]	[5]	[6]
VARIABLES						
RBL adoption	-0.933*** (0.245)	-0.562*** (0.192)	-1.058*** (0.267)	-0.942*** (0.157)	-1.209*** (0.291)	-0.510** (0.226)
Constant	8.509*** (0.147)	8.574*** (0.294)	11.886*** (1.557)	9.063*** (0.863)	11.011*** (1.586)	8.357*** (0.884)
Covariates	No	No	Yes	Yes	Yes	Yes
Country/FE	No	Yes	No	Yes	No	Yes
Year/FE	No	Yes	No	No	Yes	Yes
Observations	306	306	306	306	306	306
R-squared	0.047	0.904	0.335	0.958	0.381	0.961

Robust standard errors in parentheses.*** p<0.01, ** p<0.05, * p<0.1

8 Heterogeneity

8.1 Effect by type of interest rate, nature of RBL and lenders

In this section, we analyze the heterogeneous effects of resource-backed loans with respect to either the different interest rates applied or creditor countries. NRGi's RBL database indicates that China is the largest lender of RBLs, ahead of countries such as Russia. We subdivide lenders into two types: RBLs granted by China and then group the lenders (namely, Russia, Nigeria, Korea, etc.). Besides, the NRGi report that 7 of the resource-backed loans have fixed rates, while 10 have variable interest rates. In particular, Chinese strategic banks offered fixed interest rates as low as 0.25% on export loans, the China Eximbank loan to the Republic of Congo in 2009 being a good example. On their side, Niger and Zimbabwe benefited from interest rates as high as %. It should be noted that China Eximbank offered all fixed interest RBL transactions by China; in addition, floating interest rates generally use the London Interbank Offered Rate (LIBOR) as a reference rate, with rates ranging from Libor (1.0 percent) to Libor (2.95 percent). On their side, Korean loans to the DRC had the lowest weighted interest rate among all loans granted, at 0.01% (see [Mihalyi et al., 2020](#)). Table 9 reports the results according to: it turns out that resource-backed loans reduce the sovereign credit rating regardless of the type of interest rate (panel A columns [1]-[2]) or the type of lender (panel A columns [3]-[4]), with

a high proportion of RBLs granted by other lenders, such as Russia, Nigeria, and Korea. Finally, in panel B columns [1]-[3], we disaggregate the RBL variable into three transaction types following [Coulibaly \(2024\)](#). The results show that oil and mineral RBLs have a negative and statistically significant impact on sovereign credit ratings, while tobacco and cocoa RBLs do not have a statistically significant effect.

Table 9: Resource-backed loans and SCR: heterogeneity results

Panel A: Type of interest rates	[1]	[2]	[3]	[4]
Floating interest rate	-2.0158** (0.8947)			
Fixed interest rate		-2.5036*** (0.7291)		
China's RBL			-0.9233** (0.4678)	
Others lenders RBL				-3.3284*** (0.7486)
Constant	0.1450 (4.3357)	-2.0224 (3.9469)	-0.8388 (4.2398)	-2.0563 (3.8549)
Covariates	Yes	Yes	Yes	Yes
Country/FE & Year/FE	Yes	Yes	Yes	Yes
Observations	603	603	603	603
R-squared	0.8333	0.8384	0.8327	0.8415
Panel B: Type of collateral	[1]	[2]	[3]	
Oil RBLs	-1.2354* (0.6632)			
Mineral RBLs		-1.7755*** (0.4993)		
Tobacco and cocoa RBLs			-1.6885 (1.1859)	
Constant	0.0261 (3.7150)	-1.7765 (4.0414)	-1.5965 (3.9972)	
Covariates	Yes	Yes	Yes	
Country/FE & Year/FE	Yes	Yes	Yes	
Observations	639	639	639	
R-squared	0.8334	0.8324	0.8301	

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

8.2 The effect of resource-backed loan over time

This section tests whether the adoption of resource-backed loans has a persistent or temporary effect. To achieve this, we estimate the impact of a resource-backed loan over five years after its adoption. The results in Table 10 show that the effect of the resource-backed loan persists. This effect appears as soon as a resource-backed loan is adopted and increases over time to approach the (average) magnitude of the reference effect five years after the policy is adopted.

Table 10: Resource-backed loans and SCR: perspective time

VARIABLES	[1]	[2]	[3]	[4]	[5]	[6]
RBL (t0)	-2.4193*** (0.4979)					
RBL (t+1)		-2.4193*** (0.4979)				
RBL (t+2)			-2.5885*** (0.5949)			
RBL (t+3)				-2.9209*** (0.7017)		
RBL (t+4)					-2.7660*** (0.8184)	
RBL (t+5)						-2.8804*** (0.7113)
Constant	0.1353 (4.0087)	0.1353 (4.0087)	0.2750 (4.0569)	0.3469 (4.3515)	-2.7765 (5.9408)	-11.0683** (5.3741)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE & Year/FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	602	602	565	528	491	454
R-squared	0.8544	0.8544	0.8537	0.8531	0.8536	0.8658

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1 Note: compared with Table 3, the observations linked to the treatment variable differ. The first column constructs a new treatment variable by deleting all observations following the year of resource-backed loan adoption. Columns [2]-[6] consider only observations for 1, 2, 3, 4, and 5 years after the year of resource-backed loan adoption, respectively.

8.3 The role of structural factors

Our starting point is the idea that developing countries have different macroeconomic and political conditions (see, e.g. [Acemoglu et al., 2019](#); [Acemoglu et al., 2014](#); [Coulibaly, 2024](#); [Balima et al., 2017](#); [Minea and Tapsoba, 2014](#)). These structural and institutional differences may, to some extent, shape the effect of resource-backed loans on sovereign credit ratings. Consequently, this section examines the heterogeneity effect of resource-backed loans according to five macroeconomic characteristics (public debt, tax revenue, natural resources rents, total investment, government expenditure) and six institutional characteristics (corruption, government effectiveness, control of corruption, political stability, rule of law, regulatory quality).

First, we analyze the effectiveness of resource-backed loans in terms of public debt, tax revenue, resource rent, total investment, and public spending. Given the link with the variable SCR or RBL (see, e.g. [Coulibaly, 2024](#); [Reusens and Croux, 2017](#); [Mellios and Paget-Blanc, 2006](#); [Baum et al., 2016](#); [Paudyn, 2013](#); [Sawadogo, 2020](#)), we hypothesize that the adoption of resource-backed loans may have a greater effect in situations characterized by high public debt, low tax revenue mobilization, high dependence on resource rent, low level of public investment, and debt-to-GDP ratio are expected to have a lower sovereign credit rating ([Reusens and Croux, 2017](#)). Therefore, managing public debt levels is critical to take advantage of resource-backed loans. Debt control and sound public finances are key factors in mobilizing financial resources in developing countries ([Reinhart et al., 2003](#); [Reinhart and Rogoff, 2010](#)), since unsustainable debt limits the capacity to access financial markets ([Sawadogo, 2020](#); [Coulibaly, 2024](#)) and prevents it from financing sustainable development policies.

The fiscal behavior of resource-rich developing countries fundamentally differs from that of countries without natural resources. As highlighted in the economic literature on the curse of resources (see e.g., [Williams, 2011](#); [Sachs and Warner, 1995](#); [James and Aadland, 2011](#); [Papyrakis and Gerlagh, 2007](#); [Sachs and Warner, 1999](#); [Sachs and Warner, 2001](#); [Cheng et al., 2021](#)), high levels of resource rent can trigger fiscal policies that impede economic growth. In such a context, the adoption of resource-backed loans plunges these countries into a situation of fiscal procyclicality, reducing fiscal policy credibility, and simply raising borrowing costs. To test these hypotheses, we include an interaction between each of these variables and the resource-backed loan. Consequently, a statistically significant interaction indicates the presence of heterogeneity. The results of [1]-[5] in Table 11 support our hypothesis. The impact of resource-backed loans on sovereign credit ratings is amplified when other macroeconomic factors are considered. This suggests that the effects of resource-backed loans on sovereign credit ratings are sensitive, to some extent, to macroeconomic conditions.

Second, and from a political and institutional perspective, we consider corruption, government effectiveness, control of corruption, political stability, rule of law, and regulatory quality. The results reported in columns [6]-[11] show that there are two important effects to analyze. First, the interaction term between RBLs and corruption has a negative effect on can be intuitively explained by the fact that countries characterized by a high level of corruption are more likely to sign up to the RBL on a discretionary basis, with no records in the official debt statistics. Under these conditions, RBLs can worsen the viability of the public debt burden and discourage traditional lenders from granting debt to the country in question. However, RBLs would be highly effective and would increase sovereign credit rating in the case of effective government effectiveness, corruption control, political stability, rule of law, and regulatory quality.

Table 11: Resource-backed loans and SCR: structural factors

VARIABLES	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]
RBL*Public debt	-0.0171*** (0.0056)										
RBL*Tax revenue		-0.1580*** (0.0456)									
RBL*Natural resources rents			-0.1209*** (0.0185)								
RBL*Total investment				-0.0349* (0.0179)							
RBL*Government expenditure					-0.1363*** (0.0265)						
RBL*Corruption						-1.2932*** (0.3031)					
RBL*Government effectiveness							2.1482*** (0.4152)				
RBL*Control of corruption								2.0694*** (0.4340)			
RBL*Political stability									1.2465*** (0.4448)		
RBL*Rule of law										2.5282*** (0.4671)	
RBL*Regulatory quality											1.9450*** (0.3908)
Constant	-1.0580 (3.9526)	-0.6759 (3.5488)	-3.4904 (3.1131)	-1.5360 (4.4557)	-4.1123 (3.7403)	-4.3695 (3.0883)	-1.2885 (3.6910)	0.2500 (3.8258)	-0.4383 (3.9279)	-0.8598 (3.6531)	-0.6586 (3.6864)
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country/FE & Year/FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	639	340	639	604	445	420	639	639	639	639	639
R-squared	0.8323	0.8789	0.8503	0.8188	0.8115	0.8706	0.8418	0.8394	0.8340	0.8423	0.8400

Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

9 Conclusion

Over the past few decades, the issue of sovereign credit ratings has become central to the quest of sub-Saharan African countries for broader and less costly access to international capital markets. In a context marked by an increasing dependence on external debt, the emergence of eurobonds, and persistent macroeconomic vulnerabilities, sovereign ratings

play a decisive role in the ability of states to finance their development. Yet many countries in the region continue to have low or speculative ratings, often affected by chronic budget deficits, weak economic diversification, and fragile institutions. This is the backdrop to the growing use of resource-backed loans (RBLs), an alternative financing mechanism that raises questions about its sustainability and implications for the financial credibility of governments.

This study analyzed the effect of RBLs on sovereign ratings using a sample of 45 sub-Saharan African countries over the period 1990-2019. Combining rigorous econometric approaches and various robustness tests, the results highlight a significant negative effect of the use of RBLs on sovereign credit ratings robust to the use of an alternative dependent variable from the World Bank, the sample composition (by excluding countries in conflict or net oil exporters), or a placebo test. The analysis suggests that RBLs, while attractive in the short term for financing infrastructure projects or filling budget deficits, are perceived by rating agencies as opaque and vulnerable to commodity price volatility.

These results call for a broader reflection on prudent debt management in developing countries, particularly those with declining access to concessional financing. Promoting contractual transparency, integrating RBLs into official debt statistics, and strengthening institutional frameworks for fiscal management appear to be priorities for improving sovereign credibility. More broadly, this study contributes to the literature by empirically documenting a channel through which certain innovative financial instruments can weaken, rather than strengthen, the macro-fiscal position of low-income countries.

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Appendix

Table A1: Descriptive statistics

Variable	Obs	Mean	Std. Dev.	Min	Max
SCR	792	3.239	4.556	0	16
RBL	1350	0.086	0.28	0	1
Lag GDP per capita	1227	1.417	7.365	-48.392	140.48
Lag Inflation rate	1109	51.023	746.223	-11.686	23773.131
Commodity price	1147	95.644	8.852	45.012	106.913
Political stability	1075	0.472	0.156	0	0.791
Resource depletion	1241	9.498	11.608	0	102.891
Lag Public debt	960	66.734	66.151	0	600.117
Financial openness	1261	-0.677	1.221	-1.927	2.311
Total reserves	823	3.681	3.578	0.002	26.729
Resources rich	720	0.375	0.484	0	1
Exchange rate dummy	1167	6.351	4.763	1	15
Government stability	870	7.59	2.114	0.667	11.583
Control corruption	1075	0.275	0.136	0	0.706
Debt default	270	0.367	0.483	0	1
Inflation crises	269	0.249	0.433	0	1

Table A2: Correlation matrix

Variables	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]
SCR	1.000															
RBL	-0.082	1.000														
Lag GDP per capita	0.075	-0.006	1.000													
Lag Inflation rate	-0.088	-0.015	-0.061	1.000												
Commodity prices	0.073	0.049	0.124	-0.334	1.000											
Political stability	0.402	-0.073	0.062	-0.116	0.090	1.000										
Resource depletion	-0.258	0.131	0.107	0.123	-0.447	-0.236	1.000									
Lag Public debt	-0.285	-0.060	-0.079	0.198	-0.175	-0.115	0.111	1.000								
Financial openness	0.269	-0.092	0.017	-0.048	0.010	0.298	-0.162	0.123	1.000							
Total reserves	0.447	-0.089	0.016	-0.063	0.138	0.254	-0.193	-0.269	0.168	1.000						
Resources rich	0.035	0.214	0.045	-0.052	-0.502	0.033	0.390	0.053	0.037	0.163	1.000					
Exchange rate dummy	0.165	-0.057	-0.092	0.109	-0.199	-0.092	0.023	0.174	0.177	-0.028	-0.063	1.000				
Government stability	0.033	0.053	0.182	-0.023	0.073	0.258	0.031	0.015	0.103	0.237	0.056	-0.172	1.000			
Control of corruption	0.585	-0.162	0.032	-0.066	0.231	0.628	-0.440	-0.059	0.345	0.324	-0.211	0.165	0.170	1.000		
Debt default	-0.664	-0.005	-0.291	0.183	-0.222	-0.553	0.098	0.245	-0.301	-0.238	-0.201	-0.382	-0.181	-0.525	1.000	
Inflation crises	-0.285	-0.050	-0.223	0.303	-0.514	-0.101	0.422	0.339	-0.195	-0.175	0.266	0.519	-0.104	-0.212	0.153	1.000

Table A3: Conversion of sovereign credit ratings

Moody's	S&P	Fitch	Description	Rating
Aaa	AAA	AAA	Prime	21
Aa1	AA+	AA+	High grade	20
Aa2	AA	AA		19
Aa3	AA-	AA-		18
A1	A+	A+	Upper medium grade	17
A2	A	A		16
A3	A-	A-		15
Baa1	BBB+	BBB+	Lower medium grade	14
Baa2	BBB	BBB		13
Baa3	BBB-	BBB-		12
Ba1	BB+	BB+	Non-investment grade speculative	11
Ba2	BB	BB		10
Ba3	BB-	BB-		9
B1	B+	B+	Highly speculative	8
B2	B	B		7
B3	B-	B-		6
Caa1	CCC+	CCC	Substantial risks	5
Caa2	CCC	CCC		4
Caa3	CCC-	CCC (low)		3
Ca	CC	CC		2
C	C	C		1
D	/	DDD		
DD	D		In default	-1
D				
Not rated				0

Table A4: Definition and sources of data

Variables	Definition	Sources
1. Dependent variable		
Sovereign credit rating	Continuous	Author's calculation
Sovereign credit rating	Foreign currency long-term sovereign debt ratings (index ranging from 1 to 21, higher value means better rating)	Kose et al. (2017)
2. Treatment variable		
Resource-backed loans (RBLs)	Takes value 1 if country i at period t has signed a resource-backed loans	Mihalayi et al. (2020) ; Mihalayi et al. (2022) ; Coulibaly (2024) ; Coulibaly (2025)
Oil resource-backed loans	Takes value 1 if country i at period t has signed a loans on oil	Mihalayi et al. (2020) ; Mihalayi et al. (2022) ; Coulibaly (2024) ; Coulibaly (2025)
Minerals resource-backed loans	Takes value 1 if country i at period t has signed a loans on minerals	Mihalayi et al. (2020) ; Mihalayi et al. (2022) ; Coulibaly (2024) ; Coulibaly (2025)
Other resource-backed loans	Takes value 1 if country i at period t has signed a loans on tobacco and cocoa	Mihalayi et al. (2020) ; Mihalayi et al. (2022) ; Coulibaly (2024) ; Coulibaly (2025)
3. Baseline model control variables or covariates		
Inflation rate	Continuous	WDI
Resource depletion	Continuous	WDI
GDP per capita	Continuous	WDI
Commodity price	Continuous	IMF
Public debt	Continuous	WDI
Political stability	Continuous	ICRG
4. Additional control variables		
Capital openness	Continuous	Chinn and Ito (2006)
Total reserve	Continuous	WDI
Resource-rich	Dummy	IMF
Exchange regime	Continuous	Chinn and Ito (2006)
Governance stability	Continuous	ICRG
Control of corruption	Continuous	ICRG
Payment defaults	Continuous	Reinhart and Rogoff (2009)
Inflation crisis	Dummy	Reinhart and Rogoff (2009)
Tax revenue	Continuous	WDI
Natural resources rents	Continuous	WDI
Total investment	Continuous	IMF
Government expenditure	Continuous	WDI
Corruption	Continuous	ICRG
Government effectiveness	Continuous	ICRG
Control of corruption	Continuous	ICRG
Rule of law	Continuous	ICRG
Regulatory quality	Continuous	WDI